Wei Bai

Member of Technical Staff at Microsoft AI

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About

Networking researcher and engineer with 10+ years across academia and industry, spanning the full stack of AI/data center networking — from GPU collective communication and RDMA transport to NIC and switch architecture. I have co-architected RDMA transports running on hundreds of thousands of GPUs in production (OpenAI, Microsoft), helped build and operate one of the world's largest RoCE deployments (spanning up to 100 km and carrying over 70% of Azure traffic), published at top-tier academic conferences such as SIGCOMM and NSDI, and contributed to open-source projects and open industry standards.

Work Experience

Microsoft *Redmond, USA*

Member of Technical Staff, Microsoft AI *Feb. 2026 – Present*

- Designing networking, RDMA transport, and collective communication mechanisms for large-scale AI clusters.

NVIDIA *Redmond, USA*

Principal Software Research Architect, Networking Software and System Architecture Group *Jan. 2024 – Feb. 2026*

- **Spectrum-X:** Contributed to Spectrum-X, NVIDIA's Ethernet networking platform for AI, focusing on long-distance DC-to-DC and region-to-region networking scenarios.
- **MRC:** Served as a core architect of Multipath Reliable Connection (MRC), a next-generation RDMA transport protocol that distributes a single RDMA connection across multiple network paths. Brought MRC to production across hundreds of thousands of GPUs at OpenAI and Microsoft, and provided hands-on deployment support. MRC has since been published as an Open Compute Project (OCP) open specification.

Microsoft *Redmond, USA & Beijing, China*

Senior Researcher, Microsoft Research Redmond (MSR), Aug. 2019 – Jan. 2024
Researcher, Microsoft Research Asia (MSRA), Jul. 2017 – Aug. 2019 *Jul. 2017 – Jan. 2024*

- **RDMA for Azure:** Core contributor to one of the world's largest RoCE (RDMA over Converged Ethernet) deployments at Microsoft Azure, spanning data centers up to 100 km apart and carrying over 70% of Azure traffic. Worked across four generations of RDMA NICs and more than ten switch ASICs from Broadcom, Cisco, and NVIDIA, covering congestion control, switch buffer optimization, and end-to-end NIC/switch qualification. The resulting RoCE network underpins Azure Storage and enabled the inter-datacenter training of GPT-4.5.
- **SONiC:** Contributed to SONiC (Software for Open Networking in the Cloud), an open-source network operating system now deployed across major cloud and enterprise networks. Delivered disaggregated chassis support, QoS/RDMA features, and a programmable-hardware-based test system using platforms such as Keysight.
- **Next-Generation Cloud and AI Networking Research:** Led research on the next generation of RDMA and AI networking, covering the full spectrum from transport protocol design and NIC testing to switch buffer management and RDMA performance isolation. Uncovered multiple previously unknown bugs in production RDMA NICs through systematic testing. This work directly influenced the design of next-generation NIC and switch architectures.

Education

The Hong Kong University of Science and Technology (HKUST) *Hong Kong SAR, China*

PH.D. IN COMPUTER SCIENCE AND ENGINEERING *2013 – 2017*

- Advisor: Prof. Kai Chen
- Thesis: "Congestion Control Mechanisms for Data Center Networks"
- Microsoft Research Asia Fellowship (2015)
- USENIX NSDI Student Grant (2015, 2016)
- HKUST Research Travel Grant (2016, 2017)

Shanghai Jiao Tong University (SJTU) *Shanghai, China*

B.E. IN INFORMATION SECURITY *2009 – 2013*

- Outstanding Bachelor's Thesis (Top 1%)
- First Prize, Information Security Contest, National Undergraduate Electronic Design Contest (2012)

Publications

CONFERENCE PUBLICATIONS

[C32] Federico De Marchi, Jialong Li, Ying Zhang, **Wei Bai**, Yiting Xia, "Unlocking Superior Performance in Reconfigurable Data Center Networks with Credit-Based Transport", in *Proceedings of the ACM SIGCOMM 2025 Conference* (**SIGCOMM**), 2025.

[C31] Jialong Li, Haotian Gong, Federico De Marchi, Aoyu Gong, Yiming Lei, **Wei Bai**, Yiting Xia, "Uniform-Cost Multi-Path Routing for Reconfigurable Data Center Networks", in *Proceedings of the ACM SIGCOMM 2024 Conference* (**SIGCOMM**), Sydney, Australia, 2024.

[C30] Jiaqi Lou*, Xinhao Kong*, Jinghan Huang, **Wei Bai**, Nam Sung Kim, Danyang Zhuo, "Hardware-assisted RDMA Performance Isolation for Public Clouds", in *Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), 2024.

[C29] Vamsi Addanki, **Wei Bai**, Stefan Schmid, Maria Apostolaki, "Reverie: Low Pass Filter-Based Switch Buffer Sharing for Datacenters with RDMA and TCP Traffic", in *Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), 2024.

[C28] Hao Wang, Han Tian, Jingrong Chen, Xinchen Wan, Jiacheng Xia, Gaoxiong Zeng, **Wei Bai**, Junchen Jiang, Yong Wang, Kai Chen, "Towards Domain-Specific Network Transport for Distributed DNN Training", in *Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), 2024.

[C27] Zhuolong Yu, Bowen Su, **Wei Bai**, Shachar Raindel, Vladimir Braverman, Xin Jin, "Understanding the Micro-Behaviors of Hardware Offloaded Network Stacks with Lumina", in *Proceedings of the ACM SIGCOMM 2023 Conference* (**SIGCOMM**), New York City, New York, September 10-14, 2023.

[C26] Hwijoon Lim, Jaehong Kim, Inho Cho, Keon Jang, **Wei Bai**, Dongsu Han, "FlexPass: A Case for Flexible Credit-based Transport for Datacenter Networks", in *Proceedings of the 18th European Conference on Computer Systems* (**EuroSys**), Rome, Italy, May 8-12 2023.

[C25] Wei Bai, Shanim Sainul Abdeen, Ankit Agrawal, Krishan Kumar Attre, Paramvir Bahl, Ameya Bhagat, Gowri Bhaskara, Tanya Brokhman, Lei Cao, Ahmad Cheema, Rebecca Chow, Jeff Cohen, Mahmoud Elhaddad, Vivek Ette, Igal Figlin, Daniel Firestone, Mathew George, Ilya German, Lakhmeet Ghai, Eric Green, Albert Greenberg, Manish Gupta, Randy Haagens, Matthew Hendel, Ridwan Howlader, Neetha John, Julia Johnstone, Tom Jolly, Greg Kramer, David Kruse, Ankit Kumar, Erica Lan, Ivan Lee, Avi Levy, Marina Lipshteyn, Xin Liu, Chen Liu, Guohan Lu, Yuemin Lu, Xiakun Lu, Vadim Makhervaks, Ulad Malashanka, David A. Maltz, Ilias Marinos, Rohan Mehta, Sharda Murthi, Anup Namdhari, Aaron Ogus, Jitendra Padhye, Madhav Pandya, Douglas Phillips, Adrian Power, Suraj Puri, Shachar Raindel, Jordan Rhee, Anthony Russo, Maneesh Sah, Ali Sheriff, Chris Sparacino, Ashutosh Srivastava, Weixiang Sun, Nick Swanson, Fuhou Tian, Lukasz Tomczyk, Vamsi Vadlamuri, Alec Wolman, Ying Xie, Joyce Yom, Lihua Yuan, Yanzhao Zhang, Brian Zill, "Empowering Azure Storage with RDMA", in *Proceedings of the 20th USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), Boston, Massachusetts, April 17-19, 2023 (Operational Systems Track).

[C24] Xinhao Kong, Jingrong Chen, **Wei Bai**, Yechen Xu, Mahmoud Elhaddad, Shachar Raindel, Jitendra Padhye, Alvin R. Lebeck, Danyang Zhuo, "Understanding RDMA Microarchitecture Resources for Performance Isolation", in *Proceedings of the 20th USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), Boston, Massachusetts, April 17-19, 2023.

[C23] Bojie Li, Gefei Zuo, **Wei Bai**, Lintao Zhang, "1Pipe: Scalable Total Order Communication in Data Center Networks", in *Proceedings of the ACM SIGCOMM 2021 Conference* (**SIGCOMM**), Virtual Conference, August 23-27, 2021.

[C22] Hwijoon Lim, **Wei Bai**, Yibo Zhu, Youngmok Jung, Dongsu Han, "Towards Timeout-less Transport in Commodity Datacenter Networks", in *Proceedings of the 16th European Conference on Computer Systems* (**EuroSys**), Virtual Conference, April 26-28, 2021.

[C21] Shuihai Hu, **Wei Bai**, Gaoxiong Zeng, Zilong Wang, Baochen Qiao, Kai Chen, Kun Tan, Yi Wang, "Aeolus: A Building Block for Proactive Transport in Datacenters", in *Proceedings of the ACM SIGCOMM 2020 Conference* (**SIGCOMM**), Virtual Conference, August 10-14, 2020.

[C20] Qun Huang, Haifeng Sun, Patrick P. C. Lee, **Wei Bai**, Feng Zhu, Yungang Bao, "OmniMon: Re-architecting Network Telemetry with Resource Efficiency and Full Accuracy", in *Proceedings of the ACM SIGCOMM 2020 Conference* (**SIGCOMM**), Virtual Conference, August 10-14, 2020.

[C19] Wei Bai, Shuihai Hu, Kai Chen, Kun Tan, Yongqiang Xiong, "One More Config is Enough: Saving (DC)TCP for High-speed Extremely Shallow-buffered Datacenters", in *Proceedings of the 39th IEEE International Conference on Computer Communications* (**INFOCOM**), Virtual Conference, July 6-9, 2020.

[C18] Junxue Zhang, **Wei Bai**, Kai Chen, "Enabling ECN for Datacenter Networks with RTT Variations", in *Proceedings of the 15th International Conference on emerging Networking EXperiments and Technologies* (**CoNEXT**), Orlando, Florida, December 9-12, 2019.

[C17] Gaoxiong Zeng, **Wei Bai**, Ge Chen, Kai Chen, Dongsu Han, Yibo Zhu, Lei Cui, "Congestion Control for Cross-Datacenter Networks", in *Proceedings of the 27th IEEE International Conference on Network Protocols* (**ICNP**), Chicago, Illinois, October 7-10, 2019.

[C16] Xiaodong Yi, Junjie Wang, Jingpu Duan, **Wei Bai**, Chuan Wu, Yongqiang Xiong, Dongsu Han, "FlowShader: a Generalized Framework for GPU-accelerated VNF Flow Processing", in *Proceedings of the 27th IEEE International Conference on Network Protocols* (**ICNP**), Chicago, Illinois, October 7-10, 2019.

[C15] Bojie Li, Tianyi Cui, Zibo Wang, **Wei Bai**, Lintao Zhang, "SocksDirect: Datacenter Sockets can be Fast and Compatible", in *Proceedings of the ACM SIGCOMM 2019 Conference* (**SIGCOMM**), Beijing, China, August 19-24, 2019.

[C14] Yang Cheng, Dan Li, Zhiyuan Guo, Binyao Jiang, Jiaxin Lin, Xi Fan, Jinkun Geng, Xinyi Yu, **Wei Bai**, Lei Qu, Ran Shu, Peng Cheng, Yongqiang Xiong, Jianping Wu, "DLBooster: Boosting End-to-End Deep Learning Workflows with Offloading Data Preprocessing Pipelines", in *Proceedings of the 48th International Conference on Parallel Processing* (**ICPP**), Kyoto, Japan, August 5-8, 2019.

[C13] Zhao Lucis Li, Mike Chieh-Jan Liang, **Wei Bai**, Qiming Zheng, Yongqiang Xiong, Guangzhong Sun, "Accelerating Rule-matching Systems with Learned Rankers", in *Proceedings of the 2019 USENIX Annual Technical Conference* (**ATC**), Renton, Washington, July 10-12, 2019 (Short Paper).

[C12] Hong Zhang, Junxue Zhang, **Wei Bai**, Kai Chen, Mosharaf Chowdhury, "Resilient Datacenter Load Balancing in the Wild", in *Proceedings of the ACM SIGCOMM 2017 Conference* (**SIGCOMM**), Los Angeles, California, August 21-25, 2017.

[C11] Ziyang Li, **Wei Bai**, Kai Chen, Dongsu Han, Yiming Zhang, Dongsheng Li, Hongfang Yu, "Rate-Aware Flow Scheduling for Commodity Data Center Networks", in *Proceedings of the 36th Annual IEEE International Conference on Computer Communications* (**INFOCOM**), Atlanta, Georgia, May 1-4, 2017.

[C10] Wei Bai, Kai Chen, Li Chen, Changhoon Kim, Haitao Wu, "Enabling ECN over Generic Packet Scheduling", in *Proceedings of the 12th International Conference on emerging Networking EXperiments and Technologies* (**CoNEXT**), Irvine, California, December 12-15, 2016.

[C9] Li Chen, Kai Chen, **Wei Bai**, Mohammad Alizadeh, "Scheduling Mix-flows in Commodity Datacenters with Karuna", in *Proceedings of the ACM SIGCOMM 2016 Conference* (**SIGCOMM**), Florianópolis, Brazil, August 22-26, 2016.

[C8] Shuihai Hu, **Wei Bai**, Kai Chen, Chen Tian, Ying Zhang, Haitao Wu, "Providing Bandwidth Guarantees, Work Conservation and Low Latency Simultaneously in the Cloud", in *Proceedings of the 35th Annual IEEE International Conference on Computer Communications* (**INFOCOM**), San Francisco, California, April 10-14, 2016.

[C7] Wei Bai, Li Chen, Kai Chen, Haitao Wu, "Enabling ECN in Multi-Service Multi-Queue Data Centers", in *Proceedings of the 13th USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), Santa Clara, California, March 16-18, 2016.

[C6] Wei Bai, Li Chen, Kai Chen, Dongsu Han, Chen Tian, Hao Wang, "Information-Agnostic Flow Scheduling for Commodity Data Centers", in *Proceedings of the 12th USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), Oakland, California, May 4-6, 2015.

[C5] Shuihai Hu, Kai Chen, Haitao Wu, **Wei Bai**, Chang Lan, Hao Wang, Hongze Zhao, Chuanxiong Guo, "Explicit Path Control in Commodity Data Centers: Design and Applications", in *Proceedings of the 12th USENIX Symposium on Networked Systems Design and Implementation* (**NSDI**), Oakland, California, May 4-6, 2015.

[C4] Yangming Zhao, Kai Chen, **Wei Bai**, Minlan Yu, Chen Tian, Yanhui Geng, Yiming Zhang, Dan Li, Sheng Wang, "RAPIER: Integrating Routing and Scheduling for Coflow-aware Data Center Networks", in *Proceedings of the 34th Annual IEEE International Conference on Computer Communications* (**INFOCOM**), Hong Kong, April 26-May 1, 2015.

[C3] Hong Zhang, Kai Chen, **Wei Bai**, Dongsu Han, Chen Tian, Hao Wang, Haibing Guan, Ming Zhang, "Guaranteeing Deadlines for Inter-Datacenter Transfers", in *Proceedings of the 10th European Conference on Computer Systems* (**EuroSys**), Bordeaux, France, April 21-24, 2015.

[C2] Wei Bai, Kai Chen, Haitao Wu, Wuwei Lan, Yangming Zhao, "PAC: Taming TCP Incast Congestion Using Proactive ACK Control", in *Proceedings of the IEEE 22nd International Conference on Network Protocols* (**ICNP**), Research Triangle, North Carolina, October 21-24, 2014.

[C1] Yang Peng, Kai Chen, Guohui Wang, **Wei Bai**, Zhiqiang Ma, Lin Gu, "HadoopWatch: A First Step Towards Comprehensive Traffic Forecasting in Cloud Computing", in *Proceedings of the 33rd Annual IEEE International Conference on Computer Communications* (**INFOCOM**), Toronto, Canada, April 27-May 2, 2014.

Workshop Publications

[W6] Xinhao Kong, Jiaqi Lou, **Wei Bai**, Nam Sung Kim, Danyang Zhuo, "Towards a Manageable Intra-Host Network", in *Proceedings of the 19th Workshop on Hot Topics in Operating Systems* (**HotOS**), Providence, Rhode Island, June 22-24, 2023.

[W5] Jiacheng Xia, Gaoxiong Zeng, Junxue Zhang, Weiyan Wang, **Wei Bai**, Junchen Jiang, Kai Chen, "Rethinking transport layer design for distributed machine learning", in *Proceedings of the 3rd Asia-Pacific Workshop on Networking* (**APNet**), Beijing, China, August 17-18, 2019.

[W4] Shuihai Hu, **Wei Bai**, Baochen Qiao, Kai Chen, Kun Tan, "Augmenting Proactive Congestion Control with Aeolus", in *Proceedings of the 2nd Asia-Pacific Workshop on Networking* (**APNet**), Beijing, China, August 2-3, 2018.

[W3] Gaoxiong Zeng, **Wei Bai**, Ge Chen, Kai Chen, Dongsu Han, Yibo Zhu, "Combining ECN and RTT for Datacenter Transport", in *Proceedings of the 1st Asia-Pacific Workshop on Networking* (**APNet**), Hong Kong, China, August 3-4, 2017.

[W2] Wei Bai, Kai Chen, Shuihai Hu, Kun Tan, Yongqiang Xiong, "Congestion Control for High-speed Extremely Shallow-buffered Datacenter Networks", in *Proceedings of the 1st Asia-Pacific Workshop on Networking* (**APNet**), Hong Kong, China, August 3-4, 2017.

[W1] Wei Bai, Li Chen, Kai Chen, Dongsu Han, Chen Tian, Weicheng Sun, "PIAS: Practical Information-Agnostic Flow Scheduling for Data Center Networks", in *Proceedings of the 13th ACM Workshop on Hot Topics in Networks* (**HotNets**), Los Angeles, California, October 27-28, 2014.

Journal Publications

[J10] Gaoxiong Zeng, **Wei Bai**, Ge Chen, Kai Chen, Dongsu Han, Yibo Zhu, Lei Cui, "Congestion Control for Cross-Datacenter Networks", in *IEEE/ACM Transactions on Networking* (**ToN**), 2022.

[J9] Shuihai Hu, Gaoxiong Zeng, **Wei Bai**, Zilong Wang, Baochen Qiao, Kai Chen, Kun Tan, Yi Wang, "Aeolus: A Building Block for Proactive Transport in Datacenter Networks", in *IEEE/ACM Transactions on Networking* (**ToN**), 2021.

[J8] Yang Cheng, Dan Li, Zhiyuan Guo, Binyao Jiang, Jinkun Geng, **Wei Bai**, Jianping Wu, Yongqiang Xiong, "Accelerating End-to-End Deep Learning Workflow With Codesign of Data Preprocessing and Scheduling", in *IEEE Transactions on Parallel and Distributed Systems* (**TPDS**), 2020.

[J7] Wei Bai, Shuihai Hu, Kai Chen, Kun Tan, Yongqiang Xiong, "One More Config is Enough: Saving (DC)TCP for High-speed Extremely Shallow-buffered Datacenters", in *IEEE/ACM Transactions on Networking* (**ToN**), 2020.

[J6] Shuihai Hu, **Wei Bai**, Kai Chen, Chen Tian, Ying Zhang, Haitao Wu, "Providing Bandwidth Guarantees, Work Conservation and Low Latency Simultaneously in the Cloud", in *IEEE Transactions on Cloud Computing* (**TCC**), 2018.

[J5] Shuhao Liu, Hong Xu, Libin Liu, **Wei Bai**, Kai Chen, Zhiping Cai, "RepNet: Cutting Latency with Flow Replication in Data Center Networks", in *IEEE Transactions on Services Computing* (**TSC**), 2018.

[J4] Wei Bai, Li Chen, Kai Chen, Dongsu Han, Chen Tian, Hao Wang, "PIAS: Practical Information-Agnostic Flow Scheduling for Commodity Data Centers", in *IEEE/ACM Transactions on Networking* (**ToN**), 2017.

[J3] Hong Zhang, Kai Chen, **Wei Bai**, Dongsu Han, Chen Tian, Hao Wang, Haibing Guan, Ming Zhang, "Guaranteeing Deadlines for Inter-Datacenter Transfers", in *IEEE/ACM Transactions on Networking* (**ToN**), 2017.

[J2] Shuihai Hu, Kai Chen, Haitao Wu, **Wei Bai**, Chang Lan, Hao Wang, Hongze Zhao, Chuanxiong Guo, "Explicit Path Control in Commodity Data Centers: Design and Applications", in *IEEE/ACM Transactions on Networking* (**ToN**), 2016.

[J1] Yang Peng, Kai Chen, Guohui Wang, **Wei Bai**, Yangming Zhao, Hao Wang, Yanhui Geng, Zhiqiang Ma, Lin Gu, "Towards Comprehensive Traffic Forecasting in Cloud Computing: Design and Application", in *IEEE/ACM Transactions on Networking* (**ToN**), 2016.

Selected Talks

2025	**SONiC for AI Networking: Looking Back and Ahead**, OCP Global Summit 2025 (with Xin Liu)
2024	**Empowering AI Networking with SONiC**, OCP Global Summit 2024 (with Guohan Lu, Xin Liu)
Sep. 2023	**RDMA at Hyperscale: Experience and Future Directions**, Louisiana State University
Jun. 2023	**RDMA at Hyperscale: Experience and Future Directions**, APNet 2023
May 2023	**Empowering Azure Storage with RDMA**, Peking University
Apr. 2023	**(Guest Lecture) Inside a Datacenter: Network Infrastructure**, University of Pittsburgh
Mar. 2023	**Empowering Azure Storage with 100×100 RDMA**, University of Washington
Nov. 2022	**(Guest Lecture) Inside a Datacenter: Hardware and Software Architecture**, Xiamen University
Oct. 2022	**RDMA Deployments at Microsoft Azure: A Researcher's Experience and View**, Intel Labs
Oct. 2022	**(Guest Lecture) Inside a Datacenter: Hardware and Software Architecture**, Boston University
Jul. 2022	**Towards Understanding the Micro-Behaviors of Hardware Offloaded Network Stacks**, Intel Switch Fabric Group
Jun. 2022	**Towards Understanding the Micro-Behaviors of Hardware Offloaded Network Stacks**, Keysight
May 2022	**RDMA at Cloud Region Scale: A Researcher's Experience and View**, Systems Seminar, Rice University
Dec. 2020	**Aeolus: A Building Block for Proactive Transport in Datacenters**, Systems and Networking Seminar, Duke University
Jan. 2019	**Build Reliable Cloud Networks with SONiC and ONE**, OCP China Technology Day
Oct. 2018	**Congestion Control Mechanisms for Data Center Networks**, HotDC 2018
Aug. 2017	**Experiments with Data Center Congestion Control Research**, APNet 2017
Dec. 2016	**Enabling ECN over Generic Packet Scheduling**, CoNEXT 2016
Mar. 2016	**Enabling ECN in Multi-Service Multi-Queue Data Centers**, NSDI 2016
May 2015	**Information-Agnostic Flow Scheduling for Commodity Data Centers**, NSDI 2015

Professional Activities

Journal Editorship: Editorial Board Member (Networking Area), Journal of Systems Research (JSys), 2021–2022.

Technical Program Committee: NSDI (2025, 2026), SIGMETRICS (2022, 2023), APNet (2019–Present), SOSR (2021, 2022), APSys (2020), ICDCS (2018).

Organizing Committee: Registration Chair, SIGCOMM 2019; Web Chair, SOSR 2021; Publicity Chair, APNet 2020; Web Chair, APNet 2018, 2019.

Reviewer: IEEE/ACM Transactions on Networking, ACM Transactions on Computer Systems, ACM Transactions on Storage, ACM Transactions on Architecture and Code Optimization, IEEE Transactions on Cloud Computing, Elsevier Computer Networks, Elsevier Future Generation Computer Systems, IEEE Transactions on Network and Service Management, IEEE Communications Letters.